Per-VM statistics: `engine/stats.py`

Turns each VM's raw 5-minute series (8,640 readings per metric) into the numbers every later step uses: percentiles, off-hours behaviour, trends, disk growth and data-quality measures. Nothing here makes a decision; `recommend.py` and `alerts.py` read these numbers.

1. What it does

Input: the six series of one VM (`cpu`, `mem`, `disk`, `iops`, `net_in`, `net_out`), plus the time helpers (`HOUR`, `WEEKDAY`, `IS_WEEKEND`) from the data window. **Output:** one dictionary of statistics per VM (`vm_stats`), plus an hourly CPU profile and a weekday × hour CPU heat-map.

Missing readings (NaN) are ignored in every calculation.

2. Helper functions

Function	Logic
`pct(x, q)`	Percentile of the non-missing values; 0.0 if nothing is left
`off_mask()`	True for every reading in the shutdown window: 20:00–08:00 on any day, plus all of Saturday and Sunday (hours are IST)
`daily_trend(x)`	Straight-line trend of the daily p95 over the last 14 days (section 4)

3. Statistics produced

CPU and memory

Field	Meaning
`cpu_avg`, `cpu_p50`, `cpu_p95`, `cpu_p99`, `cpu_max`	Average and percentiles over the whole window
`mem_avg`, `mem_p95`, `mem_p99`, `mem_max`	Same for memory
`cpu_time_above_80`	Share of readings above 80% CPU
`cv`	Coefficient of variation = standard deviation ÷ average of CPU (how erratic the VM is)

p99 and max are what rightsizing uses. A VM that is busy for only 2% of the day still shows its peak in the p99, which an average hides.

Off-hours behaviour (used for scheduling)

Field	Meaning
`offhours_cpu_avg` / `offhours_cpu_p99`	Average and p99 CPU inside the shutdown window
`offhours_busy_hours`	Hours of the day (0–23) where the off-window readings have p95 CPU above 15%. These are "night job" hours
`offhours_quiet_avg`	Average off-window CPU excluding the busy hours
`work_cpu_avg`	Average CPU 09:00–19:00 on weekdays
`hourly_cpu`	Average CPU for each hour of the day (24 values)
`heatmap_cpu`	Average CPU per weekday × hour (7 × 24), used by the what-if simulator

Network, IOPS and disk

Field	Meaning
`net_p95_mbps`	p95 of `net_in + net_out` (MB/s)
`iops_p95`	p95 of IOPS
`egress_gb_30d`	Total outbound traffic: sum of `net_out` × 300 seconds ÷ 1024 (MB/s × seconds → GB)
`disk_now`	Average disk usage over the last 6 hours (72 readings)
`disk_slope_per_day`	Slope of a straight-line fit through the last 14 days of disk usage (percentage points per day)

Data quality

Field	Meaning
`days_of_data`	Valid CPU readings ÷ 288
`missing_frac`	Share of readings missing after the VM's first valid reading. Time before the VM existed is not counted as missing

4. Trend calculation (`daily_trend`)

1. Split the last 14 days into daily blocks of 288 readings.
2. Skip any day where fewer than half the readings are valid (the VM did not exist or the agent was down).
3. If fewer than 7 days remain, return no trend.
4. For each remaining day take the p95. This removes the daily cycle so it cannot fake or

hide a trend.

5. Fit a least-squares line through the daily p95 values.
6. Return `slope` (% per day), `level` (the fitted p95 for today) and `r2` (how much of the day-to-day movement the line explains).

Used for both CPU and memory. `alerts.py` raises a trend alert when slope ⩾ 0.3 %/day and $R^2 \geqslant 0.6$.

5. Parameters

All from `config.py`: `OFF_START_HOUR` 20, `OFF_END_HOUR` 8, `OFF_WEEKENDS` True, `OFFHOURS_BUSY_P95` 15, `TREND_DAYS` 14, `POINTS_PER_DAY` 288, `STEP_MIN` 5. The 80% CPU line, the 9–19 working hours, the 14-day disk window and the 6-hour disk average are fixed in the code of `stats.py`.

6. Assumptions

- Percentiles over the whole 30-day window treat every day equally (weekends included).
- Network figures are in MB/s, so the egress conversion is a simple sum times the interval length.
- An hour counts as a "night job" hour when its off-window p95 CPU exceeds 15%.
- Time-of-day logic is in IST.

7. Limitations

1. **p99 over 30 days can miss rare events.** A monthly job might not appear in the window; `recommend.py` lowers confidence for short peaks.
2. **Max is sensitive to a single reading.** `cpu_max` is one 5-minute reading, so one noisy outlier can block a downsize later.
3. **Weekends are pooled into percentiles.** A VM busy only on weekdays gets a p99 that depends on how many weekdays are in the window.
4. **Fixed constants.** The 80% line, working hours and the disk fit window are not configurable.
5. **Simulated data.** With simulated disk and network values, the disk and network statistics are only as realistic as the generator (see `DATA_GENERATION.md`).